MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 10 · ALGEBRA II · TOPIC 2

Lesson 2-5

Completing the Square

Quadratic Functions and Equations

Mr Malek Thiab

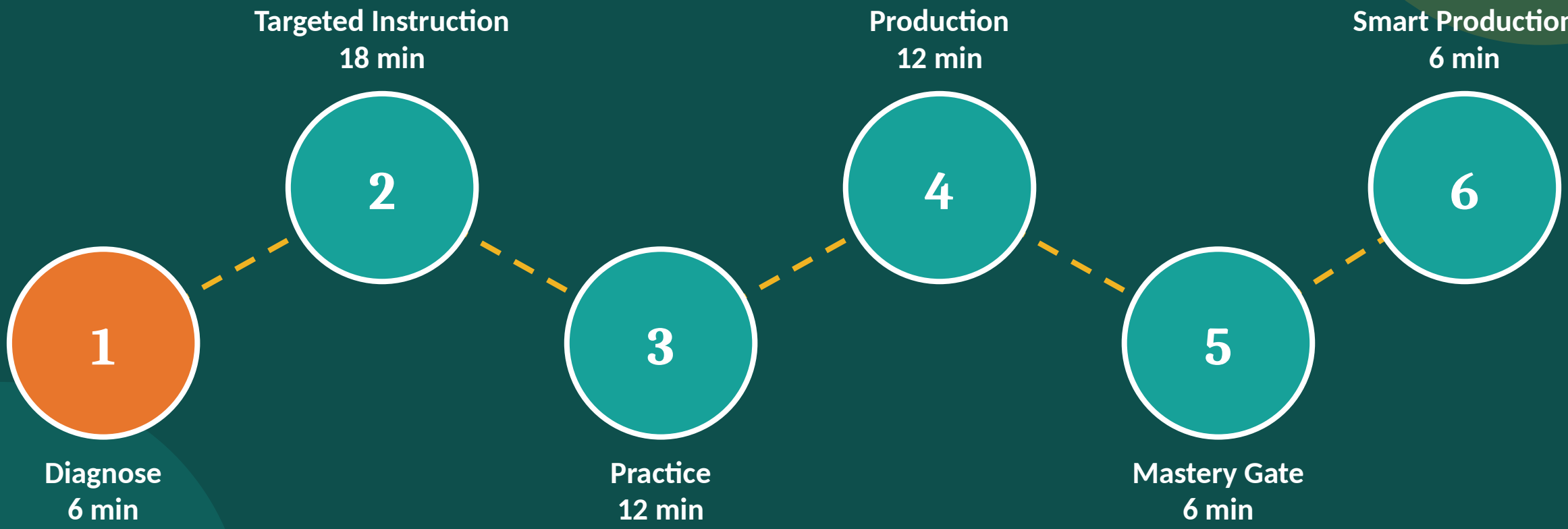
$$(x-p)^2 = q$$

Journey of 6 stations, 60 minutes

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Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Rewrite a quadratic equation into the form $(x - p)^2 = q$ by completing the square.
2. Determine the minimum or maximum value of a quadratic function using completing the square.
3. Solve real-world problems modeled by quadratic equations using completing the square.

VOCABULARY

- adjustment term
- completing the square

Standards

HSN.CN.C.7 Solve quadratic equations with real coefficients that have complex solutions.

HSA.SSE.B.3.B Complete the square in a quadratic expression to reveal the maximum or minimum value of the function it defines.

HSA.REI.B.4.A Use the method of completing the square to transform any quadratic equation in x into an equation of the form (x - p)2 = q that has the same solutions. Derive the quadratic formula from this form.

HSA.REI.B.4.B Solve quadratic equations by inspection (e.g., for x2 = 49), taking square roots, completing the square, the quadratic formula and factoring, as appropriate to the initial form of the equation. Recognize when the quadratic formula gives complex solutions and write them as a ± bi for real numbers a and b.

HSF.LE.B.6 Apply quadratic functions to physical problems, such as the motion of an object under the force of gravity.

HSF.IF.C.8.A Use the process of factoring and completing the square in a quadratic function to show zeros, extreme values, and symmetry of the graph, and interpret these in terms of a context.

STATION 1 · DIAGNOSE · 6 MIN

Start here: three quick checks

1

Expand: $(x+4)^2$

2

Factor: $x^2+10x+25$

3

Solve: $(x-3)^2=16$

Done when

All three answers are written without a calculator, and you can say why Question 3 has two solutions.

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Perfect squares and the adjustment term

$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2 \qquad \text{adjustment term} = \left(\frac{b}{2}\right)^2$$

EXAMPLE A

Complete: $x^2 + 10x + \square$

$$\left(\tfrac{10}{2}\right)^2 = 25$$

$$= (x+5)^2$$

EXAMPLE B

Complete: $x^2 - 7x + \square$

$$\left(\tfrac{-7}{2}\right)^2 = \tfrac{49}{4}$$

$$= \left(x - \tfrac{7}{2}\right)^2$$

EXAMPLE C

Complete: $x^2 + \tfrac{2}{3}x + \square$

$$\left(\tfrac{1}{3}\right)^2 = \tfrac{1}{9}$$

$$= \left(x + \tfrac{1}{3}\right)^2$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Solve by completing the square

$$x^2 + bx = c \;\Rightarrow\; \left(x + \tfrac{b}{2}\right)^2 = c + \left(\tfrac{b}{2}\right)^2$$

Add the adjustment term to BOTH sides

EXAMPLE A

Solve: $x^2 + 6x - 7 = 0$

$x^2 + 6x = 7 \;\Rightarrow\; (x+3)^2 = 16$

$x + 3 = \pm 4 \;\Rightarrow\; x = 1 \text{ or } x = -7$

EXAMPLE B

Solve: $x^2 - 8x + 5 = 0$

$x^2 - 8x = -5 \;\Rightarrow\; (x-4)^2 = 11$

$x = 4 \pm \sqrt{11}$

EXAMPLE C

Solve: $x^2 + 4x + 13 = 0$

$x^2 + 4x = -13 \;\Rightarrow\; (x+2)^2 = -9$

$x + 2 = \pm 3i \;\Rightarrow\; x = -2 \pm 3i$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

When the coefficient of x squared is not 1

$ax^2 + bx + c = 0 \Rightarrow$ divide every term by a then complete the square as before

EXAMPLE A

Solve: $2x^2 - 12x + 7 = 0$

$$x^2 - 6x = -\frac{7}{2} \Rightarrow (x-3)^2 = \frac{11}{2}$$

$$x = 3 \pm \frac{\sqrt{22}}{2}$$

EXAMPLE B

Solve: $3x^2 + 12x + 15 = 0$

$$x^2 + 4x + 5 = 0 \Rightarrow (x+2)^2 = -1$$

$$x = -2 \pm i$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Maximum and minimum values

$$y = a(x-h)^2 + k \Rightarrow \text{vertex } (h,k) \qquad a>0: \text{ minimum } k \qquad a<0: \text{ maximum } k$$

EXAMPLE A

Minimum of $f(x) = x^2 - 6x + 11$

$$= (x^2 - 6x + 9) + 11 - 9$$

$$= (x-3)^2 + 2: \text{ minimum } 2 \text{ at } x = 3$$

EXAMPLE B

Maximum of $f(x) = -2x^2 + 8x + 1$

$$= -2\left(x^2 - 4x + 4\right) + 8 + 1$$

$$= -2(x-2)^2 + 9: \text{ maximum } 9 \text{ at } x = 2$$

EXAMPLE C

Minimum of $f(x) = 3x^2 + 12x + 7$

$$= 3\left(x^2 + 4x + 4\right) - 12 + 7$$

$$= 3(x+2)^2 - 5: \text{ minimum } -5$$

STATION 3 · PRACTICE · 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

Complete: $x^2 + 12x + \square$

Solve: $x^2 - 6x - 16 = 0$

Done when: the adjustment term is added to both sides and both solutions are checked by substitution.

APPLY

Use it in context

Solve: $2x^2 + 8x - 5 = 0$

Minimum of $f(x) = x^2 + 8x + 3$

Done when: you divide by the leading coefficient first, write the vertex or the solutions in exact form, and say where the minimum occurs.

INVESTIGATE

Find out why

Why add $\left(\frac{b}{2}\right)^2$ to BOTH sides?

Complete the square on $ax^2 + bx + c = 0$

Done when: your argument keeps the equation balanced at every step, and the second item ends with the quadratic formula.

HSN.CN.C.7 · HSA.SSE.B.3.B · HSA.REI.B.4.A · HSA.REI.B.4.B · HSF.LE.B.6 · HSF.IF.C.8.A

STATION 4 · PRODUCTION · 12 MIN

Production: the Jeddah Corniche water jet

THE MODEL

On the Jeddah Corniche, a Red Sea waterfront project that supports Vision 2030 tourism, engineers programme a decorative water jet. The model height of one jet above the sea surface is $h(t) = -5t^2 + 30t + 2$ metres, where t is the time in seconds after it leaves the nozzle. These are model values, not real measurements.

$$h(t) = -5t^2 + 30t + 2 \quad \text{(metres)}$$

$$\text{Sea level: } h = 0$$

1. Complete the square to write h in the form $a(t - p)^2 + q$, and state the greatest height and the time it happens.
2. Find the two times at which the jet is at a height of 42 metres.
3. Find the time at which the jet reaches sea level, to the nearest hundredth of a second.

DONE WHEN

- Task 1 shows the factor −5 taken out and the adjustment term balanced, giving a maximum of 47 metres at 3 seconds.
- Task 2 gives two times that are the same distance either side of the time of the maximum.
- Task 3 gives only the positive time and rejects the negative root with a reason.
- Every answer is written in a full sentence with units and states that the numbers are model values.

Hint for solving: after taking out −5, add 9 inside the bracket and subtract 5 times 9 outside to keep the expression equal.

STATION 5 · MASTERY GATE · 6 MIN

Mastery Gate: three questions to pass

1

Complete: $x^2 - 10x + \square$

2

Solve: $x^2 + 8x - 9 = 0$

3

Minimum of $f(x) = x^2 + 6x + 4$

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 · SMART PRODUCTION · 6 MIN

Smart Production: make one, then check it

CREATE

1. Choose two numbers p and q and write the equation $(x - p)^2 = q$, then expand it to the form $x^2 + bx + c = 0$.
2. Swap with a partner and solve each other's equation by completing the square.
3. Check one solution by substitution and say why q decides whether the solutions are real or complex.

Done when: both equations are solved correctly, the check is written, and you can explain the sign of q.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

Solve by completing the square:

$$x^2 - 4x - 12 = 0$$

Next lesson: 2-6 The Quadratic Formula.

Exam connections

SAT

What is the minimum value of $y = x^2 - 6x + 5$?

STEPS

1 $x^2 - 6x + 5 = (x-3)^2 - 9 + 5$

2 $= (x-3)^2 - 4$

3 minimum value $= -4$

GAT

Which quantity is greater?

A: min of $x^2 - 4x + 7$ B: 2

STEPS

1 $x^2 - 4x + 7 = (x-2)^2 + 3$

2 minimum of A $= 3$

3 $3 > 2 \Rightarrow$ A is greater

SAAT

The solutions of $x^2 + 6x + 2 = 0$ are

STEPS

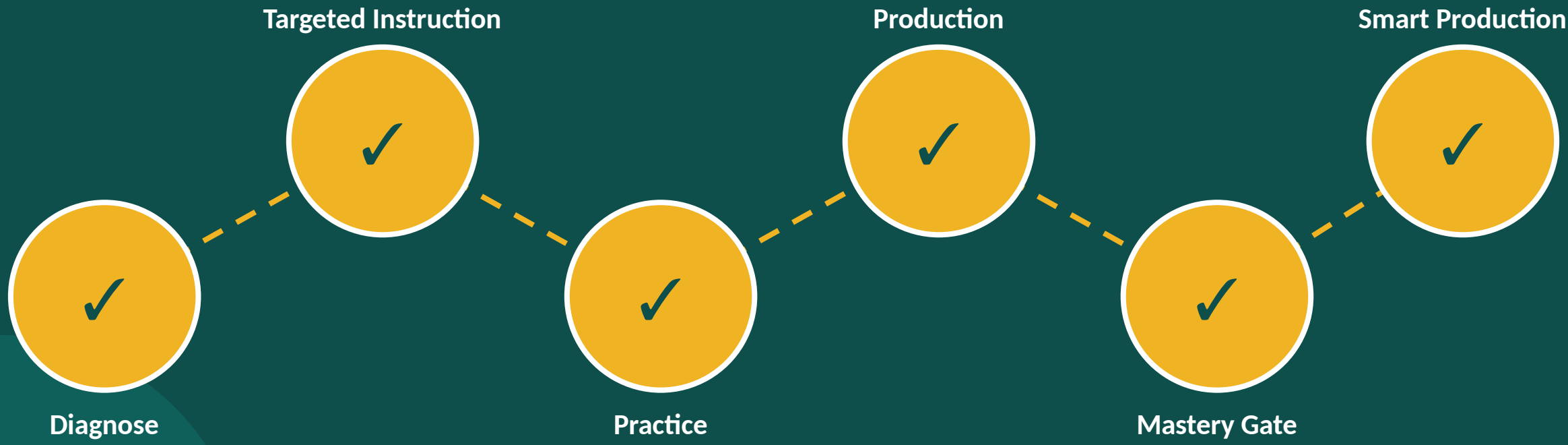
1 $x^2 + 6x = -2 \Rightarrow (x+3)^2 = 7$

2 $x + 3 = \pm\sqrt{7}$

3 $x = -3 \pm \sqrt{7}$

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Journey complete



Completing the square adds the adjustment term (b over 2) squared to both sides to make a perfect square, which solves any quadratic and reveals its maximum or minimum.